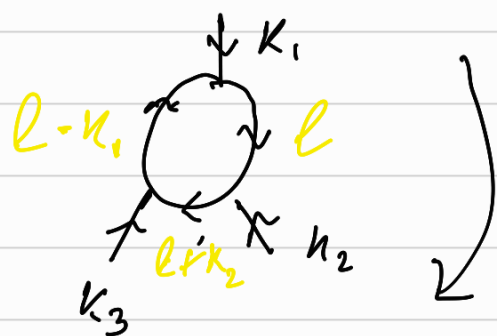


Loop Corrections to the Vertex V_n

$$iV_3(k_1, k_2, k_3) \equiv$$


$k: \text{Clockwise}$

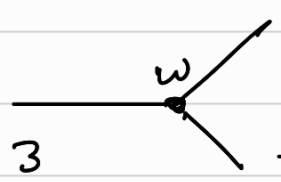
• ϕ^3 vertex

$$x + k_1 = l$$

$$\langle \Omega | \phi_1 \phi_2 \phi_3 | \Omega \rangle_c = \langle 0 | T \{ \phi_1 \phi_2 \phi_3 (\phi + g \int d^d \omega \phi_\omega^3 + g^2 \int \int + g^3 \int \int \int) \} | 0 \rangle$$

$$\mathcal{U}(g^0) = 0$$

$$\mathcal{U}(g^1) = i g \int d^d \omega \langle 0 | \underbrace{\phi_1 \phi_2 \phi_3 \phi_\omega \phi_\omega \phi_\omega}_{\text{bubble}} | 0 \rangle$$



$$= i g \int d^d \omega \Delta_{1\omega} \Delta_{2\omega} \Delta_{3\omega}$$

$$\mathcal{U}(g^2)_c \sim \langle 0 | T \{ \phi_1 \phi_2 \phi_3 \phi_\omega^3 \phi_z^3 | 0 \rangle = 0$$

$$\mathcal{O}(g^3)_c \sim (\text{Bubbles}) \times$$


Thus: $i g Z_g + (i g)^3 \left(\frac{1}{i} \right)^3 \int \frac{d^d l}{(2\pi)^3} \tilde{\Delta}(l^2) \tilde{\Delta}(l+k_2)^2 \tilde{\Delta}(l-k_1)^2 + \mathcal{O}(g^4)$

$$\text{Since } Z_g = 1 + \mathcal{O}(g^2)$$

$$Z_g^3 = 1 + \mathcal{O}(g^2)$$

* Vertex $f^n \equiv \sum_n \left(\text{all 1PI diagrams w/ } n\text{-external legs} \right)$

* For ϕ^3 :

$$iV_3(k_1, k_2, k_3) = i Z_g + (i g)^3 \frac{1}{i^3} \int \frac{d^d l}{(2\pi)^3} \tilde{\Delta}(l^2) \tilde{\Delta}(l-k_1)^2 \tilde{\Delta}(l-k_2)^2$$

$$= g \text{ (tree vertex)} + g^3 \text{ (loop vertex)} + \mathcal{O}(g^5)$$

- Note, before we were looking at loop corrections to $\langle \Omega | \phi_1 \phi_2 | \Omega \rangle$ which was a $\text{---} \bigcirc \text{---}$ corr. to $\text{---} \times \text{---}$. Here we are now looking at 'vertices' $\text{---} \text{---} \text{---}$ which have a loop correction at some $\mathcal{O}(g^n)$.

→ Evaluate $\int \frac{d^d l}{(2\pi)^d}$

$$\int \frac{d^d l}{(2\pi)^d} \tilde{\Delta}_1 \tilde{\Delta}_2 \tilde{\Delta}_3 = \int dF_3 (x_1 (l^2 - k_1^2) + x_2 (l^2 + k_2^2) + x_3 l^2 + m^2)^{-3}$$

since $\sum x_i = 1$ and

$$\int dF_3 = \frac{1}{2!} \int dx_1 dx_2 dx_3 \delta(x_1 + x_2 + x_3 - 1)$$

$$= \int [l^2 + l \cdot (x_2 k_2 - 2x_1 k_1) + x_1 k_1^2 + x_2 k_2^2 + m^2]^{-3}$$

- Note we are trying to complete the square for l and all other terms will be D
- $[l^2 - 2l(x_1 k_1 - x_2 k_2) + \dots]$

$$(l - (x_1 k_1 - x_2 k_2))^2 = l^2 - 2l(\quad) + x_1^2 k_1^2 - 2x_1 x_2 k_1 k_2 + x_2^2 k_2^2$$

add negatives of this

$$= \int [(l - (x_1 k_1 - x_2 k_2))^2 - x_1^2 k_1^2 + 2x_1 x_2 k_1 k_2 - x_2^2 k_2^2 + x_1 k_1^2 + x_2 k_2^2 + m^2]^{-3}$$

$$= \left[(\ell - (x_1 k_1 - x_2 k_2))^2 + x_1(1-x_1)k_1^2 + x_2(1-x_2)k_2^2 + 2x_1x_2k_1k_2 + m^2 \right]^{-3}$$

Define: $q \equiv \ell - (x_1 k_1 - x_2 k_2)$

$D \equiv x_1(1-x_1)k_1^2 + x_2(1-x_2)k_2^2 + 2x_1x_2k_1k_2 + m^2$

$$\tilde{\Delta}(\ell-k_1)^2 \tilde{\Delta}(\ell+k_2)^2 \tilde{\Delta}(\ell^2) = \int dF_3 [q^2 + D]^{-3}$$

Note that with $k_3^2 = k_1^2 + 2k_1k_2 + k_2^2$
 $2k_1k_2 = (k_3^2 - k_1^2 - k_2^2)$

$$\begin{aligned} D &= x_1(1-x_1)k_1^2 + x_2(1-x_2)k_2^2 + x_1x_2(k_3^2 - k_1^2 - k_2^2) + m^2 \\ &= k_1^2(x_1 - x_1^2 - x_1x_2) + k_2^2(x_2 - x_2^2 - x_1x_2) \\ &\quad + x_1x_2k_3^2 \end{aligned}$$

Since $x_1 + x_2 + x_3 = 1$

$$x_1(1 - x_1 - x_2) = x_1x_3$$

$$x_2(1 - x_2 - x_1) = x_2x_3$$

$\therefore D = x_1x_3k_1^2 + x_2x_3k_2^2 + x_1x_2k_3^2 + m^2$

$$V_3 \frac{[k_1, k_2, k_3]}{g} = Z_g - i g^2 \int dF_3 \int \frac{d^d q}{(2\pi)^d} [q^2 + D]^{-3}$$

• Now we Wick Rotate

$$q_0 = i\bar{q}_d$$

$$V_3/g = Z_g + g^2 \int dF_3 \int \frac{d^d \bar{q}}{(2\pi)^d} [\bar{q}^2 + D]^{-3}$$

• $q^d/q^6 \approx q^{d-6}$

* Diverges if $d \geq 6$, so we evaluate it for $d < 6$

$$\int \frac{d^d \bar{q}}{(2\pi)^d} [\bar{q}^2 + D]^{-3} = \frac{\Gamma(3 - \frac{1}{2}d) \Gamma(\frac{1}{2}d)}{(4\pi)^d \Gamma(3)} D^{-(3-d/2)}$$

$$= \frac{\Gamma(3-d/2)}{2(4\pi)^{d/2}} D^{-(3-d/2)}$$

• We take $[g] = \epsilon - d/2$ take $\boxed{\epsilon = d - 4}$

$\Rightarrow d = 6 - \epsilon$

$$g \rightarrow g \tilde{\mu}^{\epsilon/2}$$

$$\int = \frac{\Gamma(3 - \frac{6-\epsilon}{2})}{2(4\pi)^{6-\epsilon/2}} D^{-(3-3+\epsilon/2)} = \frac{\Gamma(\epsilon/2)}{2(4\pi)^3} \frac{(4\pi)^{\epsilon/2}}{D^{-\epsilon/2}}$$

$$V_3/g = Z_g + g^2 \frac{\tilde{\mu}^\epsilon \Gamma(\frac{\epsilon}{2})}{2(4\pi)^3} \int dF_3 \left(\frac{4\pi}{D} \right)^{\epsilon/2}$$

Define:

$$\alpha \equiv g^2 / (4\pi)^3$$

$$V_3/g = Z_g + \frac{1}{2} \alpha \Gamma(\frac{\epsilon}{2}) \int dF_3 \left(\frac{4\pi \tilde{\mu}^2}{D} \right)^{\epsilon/2}$$

$$\Gamma(\epsilon/2) = \left(\frac{2}{\epsilon} - \gamma + \mathcal{O}(\epsilon) \right)$$

$$V_3/g = Z_g + \frac{1}{2} \alpha \left(\frac{2}{\epsilon} - \gamma + \mathcal{O}(\epsilon) \right) \int dF_3 \left(1 + \frac{\epsilon}{2} \ln \left(\frac{4\pi \tilde{\mu}^2}{D} \right) \right)$$

$$= Z_g + \frac{1}{2} \alpha \left[\int dF_3 \ln \left(\frac{4\pi \tilde{\mu}^2}{D} \right) + \frac{2}{\epsilon} - \gamma \right] + \mathcal{O}(\epsilon)$$

$$\bullet \int dF_3 \ln(e^{-\gamma}) = -\gamma$$

$$= Z_g + \frac{1}{2} \alpha \left[\int dF_3 \ln \left(\frac{4\pi \tilde{\mu}^2}{e^\gamma D} \right) + \frac{2}{\epsilon} \right] + \mathcal{O}(\alpha^2)$$

Define $\mu^2 \equiv 4\pi \tilde{\mu}^2 e^{-\gamma}$

$$\therefore V_3/g = Z_g + \frac{1}{2} \alpha \left[\frac{2}{\epsilon} + \int dF_3 \ln \left(\frac{\mu^2}{D} \right) \right] + \mathcal{O}(\alpha^2)$$

* Let $Z_g = 1 + C$

$$V_3/g = 1 + C + \frac{1}{2}\alpha \left[\frac{2}{\epsilon} + \int dF_3 \ln\left(\frac{\mu^2}{D}\right) \right] + \mathcal{O}(\alpha^2)$$

$$\int dF_3 \ln\left(\frac{\mu^2 m^2}{m^2 D}\right) = 2 \ln(\mu/m) - \int dF_3 \ln\left(\frac{D}{m^2}\right)$$

$$\Rightarrow V_3/g = 1 + C + \frac{1}{2}\alpha \left[\frac{2}{\epsilon} + 2 \ln\left(\frac{\mu}{m}\right) - \int dF_3 \ln\left(\frac{D}{m^2}\right) \right]$$

$$V_3/g = 1 + C + \alpha \left[\frac{1}{\epsilon} + \ln\left(\frac{\mu}{m}\right) - \frac{1}{2} \int dF_3 \right]$$

$$V_3(k_1, k_2, k_3)/g = 1 + \left\{ \alpha \left(\frac{1}{\epsilon} + \ln\left(\frac{\mu}{m}\right) \right) + C \right\} - \frac{1}{2}\alpha \int dF_3 \ln(D/m^2)$$

Now, let:

$$C = -\alpha \left(\frac{1}{\epsilon} + \ln\left(\frac{\mu}{m}\right) + K_C \right) + \mathcal{O}(\alpha^2)$$

$K_C \in \mathbb{R}$

$$V_3(k_1, k_2, k_3)/g = 1 + \frac{1}{2}\alpha \int dF_3 \ln\left(\frac{D}{m^2}\right) - K_C \alpha + \mathcal{O}(\alpha^2)$$



* We now need a condition analogous to $\Pi(-m^2) > 0$ and $\Pi'(-m^2) = 0$ to fix the value of K_C .

• Different choices of K_C correspond to different def. of g .

$$\tau \sim V_3$$

$$\frac{d\alpha}{d\ln\mu}$$

and so the cross-section $d\sigma$ will depend on g and k_c .

* We choose $k_c = 0$

i) $V_3(0,0,0) = g$ 

Since

$$D = \beta_1 k_1^2 + \beta_2 k_2^2 + \beta_3 k_3^2 + m^2$$

$$\Rightarrow D_0 = m^2$$

ii) For $|k_1^2| \gg m^2$:

$$V_3/g \approx 1 - \frac{1}{2} \alpha \left[\ln\left(\frac{k_1^2}{m^2}\right) + \mathcal{O}(1) \right] + \mathcal{O}(\alpha^2)$$

* Compute V_4 in ϕ^4 up to $\mathcal{O}(\lambda^2)$

$$\rightarrow \int d^4x \lambda \frac{1}{4!} \phi^4 = \mathcal{L}_I$$

$$\langle \Omega | T \{ \phi_1 \phi_2 \phi_3 \phi_4 \} | \Omega \rangle_c$$

$$= (\text{free}) + \lambda i$$

$$(11 + \text{---} + \text{X}) + \lambda$$

$$+ (i\lambda)^2 \left(\begin{array}{c} \text{Diagram 1: A circle with four external lines labeled 1, 2, 3, 4. Line 1 is incoming, line 2 is outgoing, line 3 is incoming, line 4 is outgoing. A yellow 'S' is written below the circle.} \\ \text{Diagram 2: A circle with four external lines labeled 1, 2, 3, 4. Line 1 is incoming, line 2 is outgoing, line 3 is incoming, line 4 is outgoing. A yellow 't' is written below the circle.} \\ \text{Diagram 3: A circle with four external lines labeled 1, 2, 3, 4. Line 1 is incoming, line 2 is outgoing, line 3 is incoming, line 4 is outgoing. A yellow 'u' is written below the circle.} \end{array} \right)$$

• Symmetry factor $S = (1/2)$

$$(P_i) + k_1 + k_2 - l = 0$$

$$(P_s) = l - k_1 - k_2$$

$$\begin{cases} p_4 = \ell - k_1 - k_3 \\ p_2 = \ell - k_1 - k_4 \end{cases}$$

$$(i\lambda)^2 \left(\int \frac{d^d \ell}{(2\pi)^d} \Delta(\ell^2) \right) [\Delta(p_6^2) + \Delta(p_4^2) + \Delta(p_2^2)]$$

• Ignore free theory i.e.
 $(S - \mathbb{1}) = iT$

Thus, we calculate s-channel and then take $k_2 \rightarrow (k_3, k_4)$ for the last two terms:

$$(i\lambda)^2 \left(\frac{1}{i} \right)^2 \int \frac{d^d \ell}{(2\pi)^d} \left(\frac{1}{p_s^2 + m^2 - i\epsilon} \right) \left(\frac{1}{\ell^2 + m^2 - i\epsilon} \right)$$

$$\Delta(\ell^2) \Delta(p_s^2) = \int dF_2 (x p_s^2 + (1-x)\ell^2 + m^2)^{-2}$$

$$x = k_1 = k_j)^2 = x(\ell^2 - 2\ell \cdot (k_1 + k_2) + (k_1 + k_2)^2)$$

$$\Rightarrow \ell^2 + x((k_1 + k_2)^2 - 2\ell \cdot (k_1 + k_2)) + m^2$$

$$= (\ell - x(k_1 + k_2))^2 - x^2(k_1 + k_2)^2 + x(k_1 + k_2)^2 + m^2$$

$$= (\ell - x(k_1 + k_2))^2 + x(1-x)(k_1 + k_2)^2 + m^2$$

$$\boxed{\begin{aligned} q &\equiv \ell - x(k_1 + k_2) \\ D &\equiv x(1-x)(k_1 + k_2)^2 + m^2 \end{aligned}}$$

$$iV_4 = i\lambda + (i\lambda)^2 \left(\frac{1}{i} \right)^2 \int \frac{d^d q}{(2\pi)^d} \frac{1}{(q^2 + D)^2} + ()$$

* Wick rotate $q_0 \rightarrow i\bar{q}_0$

$$\int \frac{d^d q}{(2\pi)^d} \int dF \frac{1}{(q^2 + D)^2}$$

$$V_4(k_1, k_2, k_3, k_4) = \lambda + Z_\lambda \lambda^\epsilon \int (2\pi)^D \int^{D/2} \frac{d^D q}{(q^2 + D)^2}$$

$\int \sim q^{d-4}$
 • Converges for $d < 4$ and diverges for $d \geq 4$.

$$d = 4 - \epsilon \quad \leftrightarrow \quad \epsilon = 4 - d$$

$$[d^d x] = -d$$

$$[m^2] = 2$$

$$(2-d) + 2[4] = 0$$

$$[4] = \frac{1}{2}(d-2)$$

$$\therefore -d + 2(d-2) + [\lambda] = 0$$

$$d-4 + [\lambda] = 0$$

$$[\lambda] = 4-d$$

$$\lambda \rightarrow \lambda \tilde{\mu}^{\epsilon/2}$$

$$a=0, \quad b=2$$

$$\int \frac{d^d q}{(2\pi)^D} \frac{1}{(q^2 + D)^2} = \frac{\Gamma(2-d/2)}{(4\pi)^{2-d/2}} D^{-(2-d/2)}$$

$$2 - \frac{(4-\epsilon)}{2} = \epsilon/2$$

$$V_4 = \lambda \mu^{\epsilon/2} + Z_\lambda^2 \lambda^2 \frac{\mu^{\epsilon/2}}{(2\pi)^2} \Gamma\left(\frac{\epsilon}{2}\right) \int dF_2 \left(\frac{4\pi}{D}\right)^{\epsilon/2}$$

$$= V_4 = \lambda \mu^{\epsilon/2} + Z_\lambda^2 \frac{\lambda^2}{(2\pi)^2} \left(\frac{2}{\epsilon} - \gamma + d(\epsilon)\right) \int dF \left(\frac{4\pi \tilde{\mu}^2}{D}\right)^{\epsilon/2}$$

$$V_4 = \lambda \mu^{\epsilon/2} + Z_\lambda^2 \left(\frac{\lambda}{2\pi}\right)^2 \left(\frac{2}{\epsilon} - \gamma + \int dF_2 \ln\left(\frac{4\pi \tilde{\mu}^2}{D}\right)\right)$$

$$\mu^2 = 4\pi e^{-\gamma} \tilde{\mu}^2 \quad | \quad \alpha_4 = (\lambda/2\pi)^2$$

$$V_u = \lambda \mu^{\varepsilon/2} + Z_\lambda \alpha \left(\frac{2}{\varepsilon} + \int dF_2 \ln\left(\frac{\mu}{D}\right) \right)$$

$\downarrow$
 Note, this is
 $\lambda \tilde{\mu}^{\varepsilon/2} = \lambda \left(1 + \frac{2}{\varepsilon} \ln(\tilde{\mu}) + \mathcal{O}(\varepsilon^2) \right)$
 so in the limit $\varepsilon \rightarrow 0$ this is
 unchanged. $\rightarrow \lambda$

$$\begin{aligned}
 V_u &= \lambda Z_\lambda + \alpha \left(\frac{2}{\varepsilon} + 2 \ln\left(\frac{\mu}{m}\right) - \int dF_2 \ln\left(\frac{D}{m^2}\right) \right) \\
 &= \lambda Z_\lambda - \alpha \int dF_2 \ln(D/m^2) \\
 &\quad + 2\alpha \left(\frac{1}{\varepsilon} + \ln(\mu/m) \right) + \mathcal{O}(\alpha^2)
 \end{aligned}$$

$$V_u/\lambda = Z_\lambda + \alpha' \left[\frac{2}{\varepsilon} + 2 \ln\left(\frac{\mu}{m}\right) - \int dF_2 \ln\left(\frac{D}{m^2}\right) \right]$$

$$\text{Since } Z_\lambda = 1 + C \quad C \sim \mathcal{O}(\lambda^2)$$

$$V_u/\lambda = 1 + \{ \alpha' C + \mathcal{O} \} - \alpha' \int dF_2 \ln\left(\frac{D}{m^2}\right) + \mathcal{O}(\alpha'^2)$$

$$\bullet \alpha' \left(\frac{2}{\varepsilon} + 2 \ln(\mu/m) \right) + C = -\kappa_c \alpha' \in \mathbb{R}$$

$$V_u/\lambda = 1 - \kappa_c \alpha' - \alpha' \int dF_2 \ln\left(\frac{D}{m^2}\right) + \mathcal{O}(\alpha'^2)$$

$$V_u = -\lambda - \kappa_c \lambda - \frac{\lambda^2}{2(4\pi)^2} \int dF_2 \ln\left(\frac{D}{m^2}\right)$$

* Forgot to add symm factor above

$$\bullet \frac{\lambda^2}{32\pi^2} \int dF_2 \ln(D/m^2)$$

$$\rightarrow \int_0^1 dx \ln \left[\left(x(1-x)(k_1+k_2)^2 + w^2 \right) / m^2 \right]$$

$$\ln \left((x^2-x) s / m^2 + 1 \right)$$

* Assuming an-shell $S = 4m^2$

$$\ln[4(x^2 - x) + 1] \quad (2x-1)^2 = 4x^2 - 4x + 1$$

$$= 2 \ln(2x-1)$$

$$2 \int_0^1 dx \ln(2x-1) = 2 \left[(x - \frac{1}{2}) \ln(2x-1) - x \ln 2 \right]$$

$$= 2 \left[\frac{1}{2} \ln(1) - 1 - 0 \right]$$

$$= -2$$

$$V_u = -\lambda - C\lambda + \frac{x^2}{(4\pi)^2} \left[\frac{1}{\varepsilon} + \ln\left(\frac{\mu}{m}\right) + 1 \right]$$

$$S + t + u = 4m^2 \quad s_0$$

$$t + u = 0 \quad \text{and} \quad s_0$$

$$\boxed{t = 0, u = 0}$$

* Thus $D_{3,4} = m^2$

$$(V_{(p_3, p_4)})_{s_0} = \frac{i\lambda}{(4\pi)^2} \left[\frac{1}{\varepsilon} + \ln\left(\frac{\mu}{m^2}\right) \right]$$

$$V_u = -\lambda - C\lambda + \frac{\lambda^2}{(4\pi)^2} \left[\frac{3}{\varepsilon} + 3\ln\left(\frac{\mu}{m}\right) + 1 \right]$$

$$-C\lambda + \frac{\lambda^2}{(4\pi)^2} (\quad) = \lambda K_C$$

$$\therefore V_u = -\lambda + \lambda K_C + \mathcal{O}(\lambda^4)$$

Choose $K_C = 0$, s.t.

$$V_u = -\lambda + \mathcal{O}(\lambda^4)$$